

A Peek into Programming Platforms for Blockchain Development

Blockchains are no longer only about cryptocurrencies. Open source blockchain programming platforms offer security and privacy in the development of decentralised applications. Here's a list of such platforms to choose from.

Blockchain technology is bolstering security, minimising fraud and streamlining processes in various domains including wireless communications, smart cities, smart vehicular networks, finance, healthcare, supply chain, cybersecurity, and government services. The global blockchain market is growing rapidly and is expected to exceed US\$ 94

billion by 2027 as per a report from MarketsandMarkets, due to increasing interest in secure, transparent and decentralised services.

Top finance companies like JPMorgan, Visa and PayPal are spending a huge sum on blockchain based payment systems to speed up processing. Government agencies in China, the United States and the

European Union are exploring if blockchain can help with digital identity management, tax compliance and secure voting systems. Decentralised finance (DeFi), non-fungible tokens (NFTs) and enterprise blockchain applications are all part of this ecosystem. Tech giants like IBM, Microsoft and Oracle are offering blockchain-as-a-service (BaaS) so that companies can deploy blockchain solutions with negligible investment in infrastructure.

Blockchain technology is also being adopted in cybersecurity, allowing organisations to maintain records of transactions, share data securely, and detect fraud to defend themselves from the growing threat of cyber-attacks. The next few decades will see blockchain technology reshaping global digital ecosystems, offering a multitude of secure, transparent and decentralised solutions across many industries.

Blockchain-based decentralised applications (DApps) and their use cases

Blockchain is used in the deployment of DApps to ensure security and privacy. DApp based software applications are designed for execution on distributed peer-to-peer networks rather than being hosted on a centralised server. DApps differ from conventional applications in the sense that they are self-executable, censorship-resistant and utilise smart contracts to carry out pre-defined operations between parties without intermediaries. This technology



Figure 1: Blockchain and AI integrated smart vehicular communication